



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to AMBAL FUEL AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

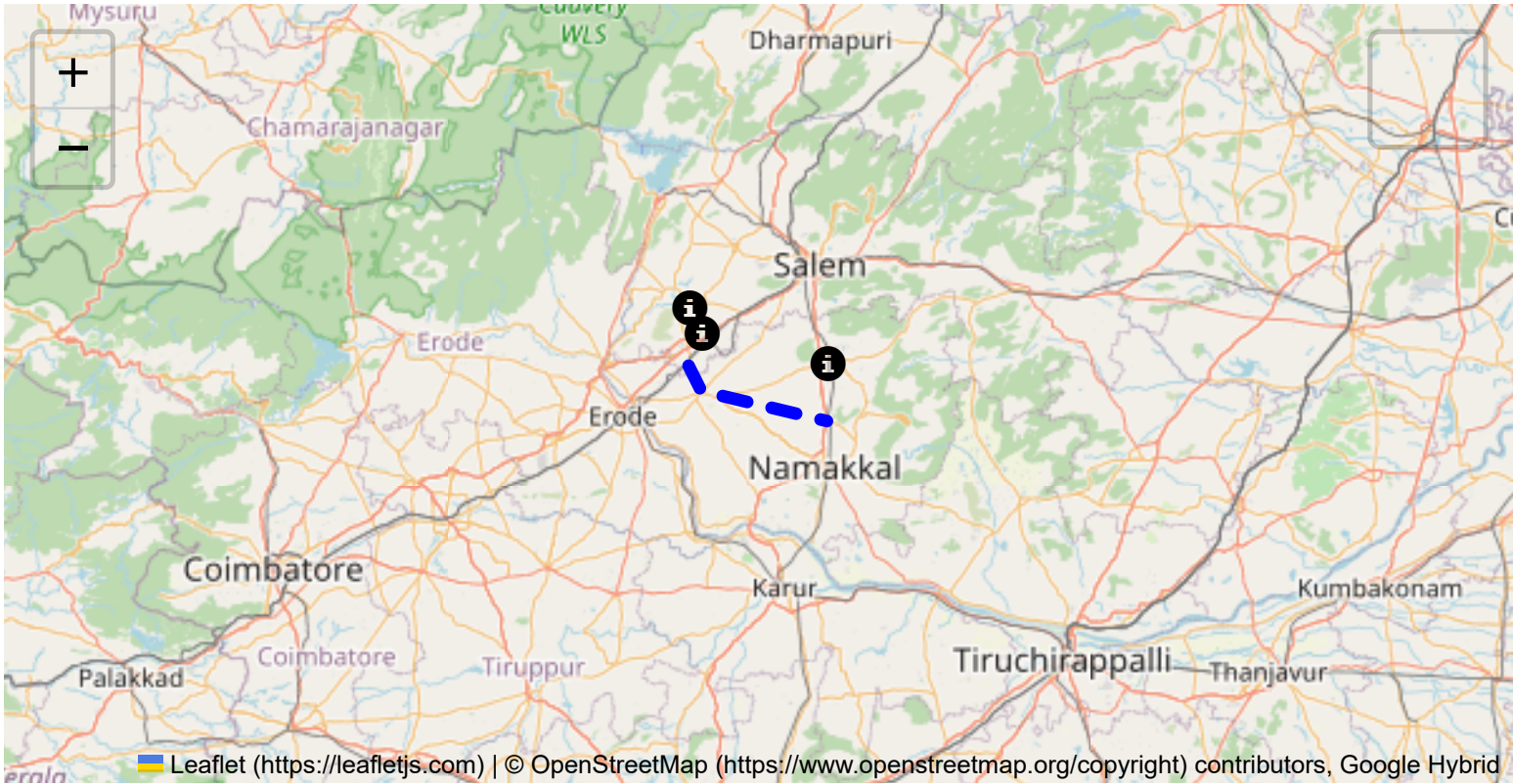
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

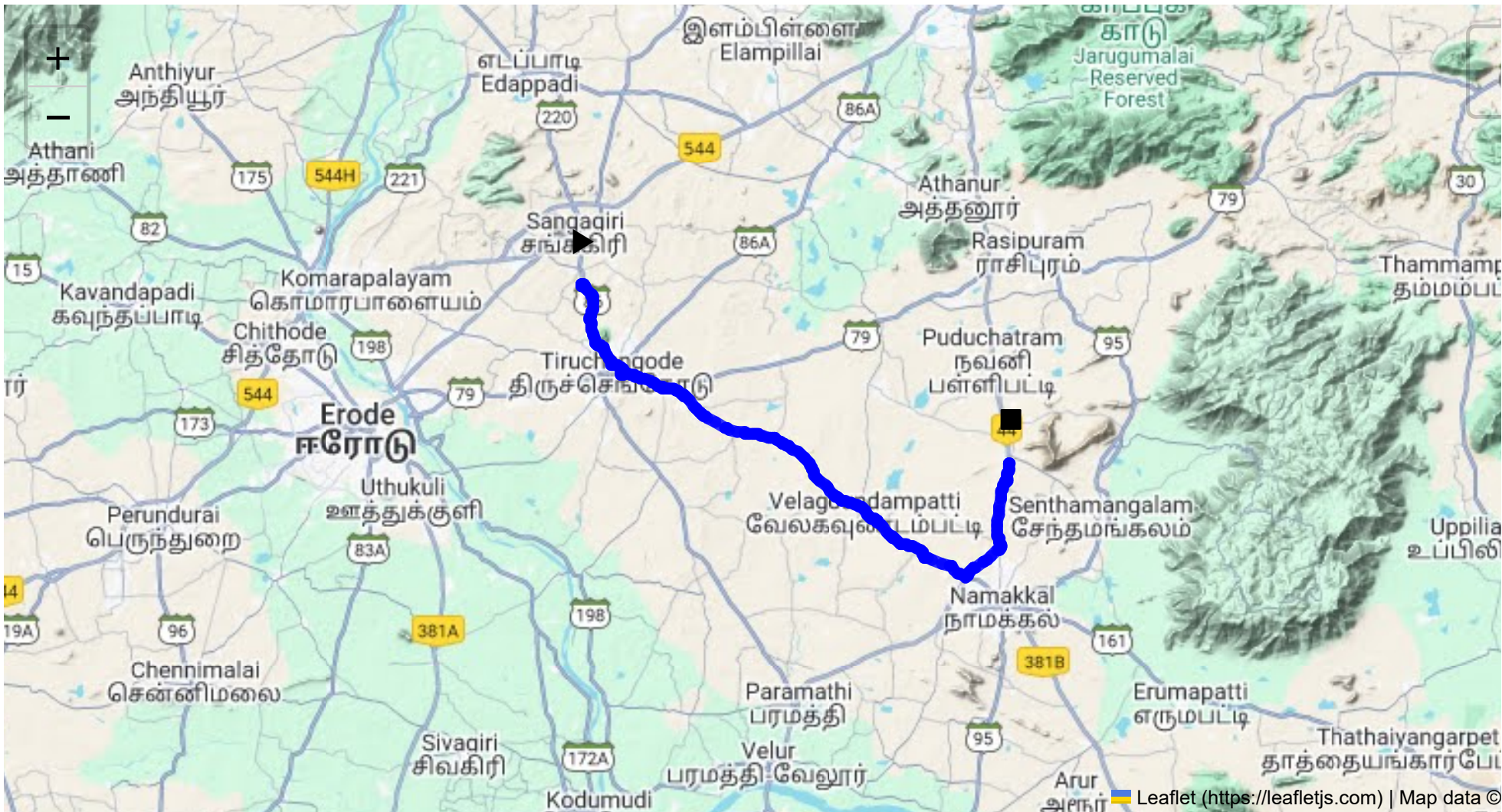
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 50.54 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.316716, 78.172074)



Welcome to the Journey Risk Management Study

Certainly! Here's a comprehensive analysis of the specified route from CVQF+23W, Sangagiri to 4/128-4 சேலம் நாமக்கல் மெயின்ரோடு, Pudanchandai, via Tiruchengode - Salem Main Rd:

1. Overview of the Route Map:

- The route spans approximately 50.54 kilometers, primarily traversing major roads such as the Tiruchengode - Salem Main Road. The journey commences from Sangagiri, passing through

Thokkavadi before reaching Pudanchandai.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- **Climate:** Predominantly tropical, with significant rainfall from June to September due to the southwest monsoon. Occasional cyclonic disturbances can occur from October to December during the northeast monsoon.
- **Hazards:** Monsoon rains may cause waterlogging and reduced visibility. Fog can also be an issue in early mornings during winter months.

3. Analysis of Traffic Patterns:

- **Peak Hours:** Morning (8 AM - 10 AM) and evening (5 PM - 7 PM) see increased traffic due to commuting.
- **Congestion-Prone Areas:**
 - Sangagiri junctions and Tiruchengode town are known for bottlenecks.
 - Entry and exit points of towns often have slow-moving traffic during peak hours.

4. Assessment of Road Quality and Infrastructure:

- **Road Condition:** Generally good but with some patches needing maintenance, especially post-monsoon. Potholes may be present.
- **Infrastructure:** Adequate signage is generally available, though some rural stretches may lack lighting.

5. Suggestions for Alternative Routes for Emergencies:

- An alternative route would be via NH44 through Salem. This main highway offers better infrastructure but can add extra kilometers to the journey.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- There's a mandate on using designated lanes for heavy vehicles in city zones.
- Transport of hazardous materials might require special permits and adherence to vehicle safety standards.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- While specific incident data may not be publicly available, regional highways can experience standard accident risks due to high traffic volumes and adverse weather.
- It's crucial to stay updated with local transport department notifications.

8. Environmental Considerations and Sensitive Areas:

- The route passes through agricultural areas; any spill or leak from vehicles can significantly impact local farming.
- Caution near water bodies to prevent contamination risks.

9. Analysis of Communication Coverage:

- Generally good network coverage in towns and along major roadways.
- Potential dead zones exist in more rural or hilly stretches of road, particularly between Tiruchengode and Thokkavadi.

10. Estimated Emergency Response Times:

- Town areas like Sangagiri and Tiruchengode have quicker access to emergency services, approximately 20-30 minutes.
- Rural stretches may experience delays of up to 45 minutes to an hour.

11. Overall Summary of Risk Assessment:

- **Risks:** Primarily related to weather, road condition variances, and minor traffic congestion. Potential for communication blackouts in rural zones.
- **Preparedness:** Drivers should be well-versed in emergency protocols, equipped with communication devices, and trained in handling hazardous materials.
- **Recommendations:** Stay alert during peak traffic hours, ensure vehicle maintenance is regularly conducted, and consider routine checks of weather and road conditions.

The analysis aims to prompt comprehensive preparations and readiness for safer transportation along this route, considering both predictable and unpredictable variables.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	U-Turn	High	11.3169758, 78.17175879999999	10 KM/Hr
2	Turn	High	11.43811, 77.87348	15 KM/Hr
3	Turn	Medium	11.43956, 77.87340	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
6	Turn	Medium	11.38346, 77.89975	30 KM/Hr
7	Blind Spot	Blind Spot	11.38188, 77.90257	10 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Turn	High	11.23961, 78.14223	15 KM/Hr
10	Turn	High	11.24549, 78.14753	15 KM/Hr
11	Turn	High	11.24537, 78.14763	15 KM/Hr
12	Blind Spot	Blind Spot	11.31698, 78.17176	10 KM/Hr
13	Turn	High	11.31697, 78.17188	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	school	Selvam Higher Secondary School	11.2555236, 78.1671989	30 km/h	Medium
16	school	Anna Nehru School	11.2580962, 78.1672223	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.3169758, 78.17175879999999



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38346, 77.89975



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.38188, 77.90257



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.23961, 78.14223



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.24549, 78.14753



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.24537, 78.14763



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.31698, 78.17176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.31697, 78.17188